

KEY DIMENSIONS - modelwheels.py TABLES				
ITEM (mm)	MAIN	NOSE	SOURCE	
Tyre (modelled)	8.50-18 Type III	17.5x5.25-6	0 owner decision 2026-09-26; 5 tyre makers / parts lists	
Part numbers	80606C-3 / 805-358-8	17060881 / 651-327-6	5 Goodgear / Michelin (parts listings)	
Ply / type / pressure	18 PR TL, 60 psi	8 PR TL, 60 psi	5 data book, POW placard, 558-62527	
From OD / section W	648 / 216	448 / 150	5 TRA 627-652 / 208-221; 6 photos 628-658	
Acta M - loaded R / defl.	279 / 41	287 / 15.5	6 main axle kept; 6 main 778-289, nose 288-210	
Contact patch / bleed R	268 / 199	139 / 78	6 (free choice 312 / 163)	
Section H / aspect	193.6 / 6.89	146.5 / 6.92	0 (TRA 8.90 / 6.92)	
Nose wheel / flange dia	254 / 6.25	152 / 3.96	5 TRA 138 28 / 6.25; 6 flange .85 / .75 in	
Between Flanges / ledge	155.8 / 34.3	127 / 22.9	5 TRA 6.25 / 5.88 in; 1.35 / 6.99 in	
Crown R / tread width	288 / 164	128 / 116	6 main, 6 nose / 6	
Shoulder R / diameter	12 / 182	16 / 97	6 / 6 (colored tangents)	
grooves: centres +- groove width x depth	36, 56 (2 pairs) 8 x 7	15, 44 (even) 6 x 5	6 head-on photos 3696, 3988, 3981	
Main centre x air / outer wheel	52 / 18 / 22	24 / 23 / 11	6 width / 6 depth	
split hub, 2 halves	split hub, 2 halves	split hub, 2 halves	5 POW 7-4-11; Goodrich 3-1543 / 3-1581	
Tie bolts n on PCO	8 on 288	4 on 110	6 hidden (main); 6 PCO, 6 count (nose)	
Web face s (skin depth)	11 (split plane)	48 (12 deep)	6 main (covered); 6 nose (challow dish, bolts seen)	
Valve / safety plug	8.77 / 1	R 65 / 3	6 position; 5 POW	
Fuelite plug	3 (board half)	-	5 POW 7-4-11; 6 position	
Hub fairing lip / face / proud	298 / 284 / 28	-	6 3986, 3985, 3858, 3988, WDX	
fairing screws / hole	5 on 288 / 855 ØR7	-	6 5 at 72 deg	
Brake	6 Ø34.1, 2 x REF, 1 wheel	-	5 POW 7-4-10; 6 3-1574-1	
3 rotors / 3 stators	-	-	5 stators / 6 rotors	
Brake discs	-	-	6 tube tips / 6	
Housing lobes / rotor dia	212 / 232	-	6 (x = 8 inward)	
Brake stack s	-120 ... -28	-	6 gear.py / 6 open bore	
axle dia / bore	52 / 45	48	6 two area (head-on photos)	
Fork area (faces) / boss	-	+88, 118 / 83	6 3981 head-on, 3986 max	
Fork arch height / arm fore-aft	-	70 / 42-52	6 position / 6 height	
kin-protector step above flange	17 x 2.5	13 x 2.0	6 flange ring width	
Flange-face bore chamfer	8	18		

PHOTO CHECK - DEVIATION CALL-OUTS (mm)				
ITEM	PHOTOS (M)	TABLE / MODEL	Δ	ACTION
Main tyre OD (modelled)	628-658	648	8	owner decision 2026-09-26
208-210-28 (user-added)	559	-	-15	MAIN TYRE CHOICE
3-D gear MAIN TYRE	648	-	0	from wheels.py
Main section width	214-224	216	8	
Main loaded radius	279-289	284.979	4	11 flat draw
Tyre R / fairing lip R	2.15-2.21	2.16	-	
retracted tyre depth	-25 (POW 1 in)	23.5	-1.9	20-38 (rit_check 5)
Main grooves / A	6.14, 6.25	6.14, 6.26	0	3-D geometry
render groove shade	-	off (geometry)	-	done
Nose tyre OD	445	448	3	
Nose loaded radius	200-210	axle 207	0	15.5 flat draw
Nose grooves / W	6.85, 6.28	6.89, 6.30	8	
Nose fork area	2	gear.py 2	0	3-D voke, round arch, saddle
Nose yoke arch height / half-span	6.85 (3081)	6.86 (p-3.1)	0	round arch (r2: square portal)
Nose area width fore-aft	42-48	42-52	0	(r2: 18-92 (padle)
Nose axle nut / lock plate	gear 1 then + tear-drop	3-D both arms	0	steel, dark
Hub fairing lip / proud	-0.58 flange R	110	0	3936 nose-hub zoom
Hub fairing dia / proud	298-388 / 25-38	298 / 28	0	6 face > 9 W/2
Main pressure	68 psi placard	gear.py info 60	8	
Main brake	6-lobe, bright cast	3-D: brake housing	0	
Main inboard tones	lobes > flange	housing 8.80 / wheel 0.24	0	
Loaded tread on the ground	Flat, across W	Flat (3-D)	0	(r2: crown r15 only

MATERIALS (model/assemble.py NAMES)				
PART	MAIN	NOSE	NOTE	
Tyre sidewall / tread / grooves	tire / _tread / _groove	see	Matte dusty / satin / dark; NO lettering	
wheel halves	wheel_half	wheel_half	weathered cast ala (main) / gloss white (nose)	
Hub fairing	hub_fairing	hub_fairing	leg-door blue (3988), a shade darker, soft coat	
Brake housing / discs, keys	brake_housing / brake_disc	-	bright satin cast ala / heat-dark steel	
bolts, valve stem	cadmium	cadmium	yellow-chrome cadmium / brass	
fairing screws, valve cap	metal / black	-	rubber-sealed cap	
axle nut, hub cap	steel	-	dark steel, nut + tear-drop lock plate (nose)	
bearing caps (nose hub ends)	chrome	chrome	polished seal rings	
arm / fork	gear_leg	gear_leg	adjacent	

LEGEND		SCALE 1:3	
	tyre (rubber) in section		
	metal in section (hatched)		
	broken view / section line		
	adjacent parts, free tyre, TRA env.		
	hidden		
	centre line, pitch circle		
	25x50-10 tyre (superadded)		
		SCALE 1:2	

- NOTES
- Drawn from the parameter tables of modelwheels.py (PTable: value + source tag + note); tyre envelopes MAIN_TYRE_ENV / NOSE_TYRE_ENV, sections, rims, fairing, brake, axle; never from the mesh. Axle positions and the gear context (trailing arm, leg, shock, nose fork) from modelgear.py, drawn as adjacent parts (phantom). Photos and their measurement working stay in the g1 ignored reticache.
 - Free-angle projection. Each assembly: SIDE view from port (geometry), FRONT view from ahead to its right (upper half in section through the axle), TOP view from above below it, for the main wheel also view C from standard (nearest) frame face at the far right. Each group side from view send on one ground line.
 - MAIN TYRE: modelled 8.50-10 Type III, OWNER DECISION 2026-09-26 tyre makers, parts lists, four photo methods: TRA OD 627-652, drawn free OD 648; Jane's / Pilatus drawing 226-50-10 (OD 558), superadded, shown in the accent colour. Axle and axle centre kept: axle WL 278 = the loaded radius, the tyre flat on the ground over a 280 mm patch and across the tread width (41 mm deflection) with a 6 mm sidewall bulge. Nose tyre loaded too: axle WL 207 (photos 200-210), 130 mm patch, 15.5 mm deflection.
 - 3-D model gear MAIN_TYRE = this envelope (R 320.0); the LD-1 leg-door scallop (R = 12.5 ± 332.0) and the round well (days WELL_5) below it, modelwheels.py main_wheel / nose_wheel revolve / extrude the profiles of the crown (tyre with the grooves, wheel halves, hub fairing, brake, valve, tie bolts) heterosymmetrically, 3D (W) compares the mesh with them.
 - Tyre section: tangent arcs - crown R to the tread half-width, shoulder round, upper sidewall (radius solved: tangent to the shoulder, max width W/2 at 0.50 H), lower sidewall with the modelled rim-protector step (17 / 13 above the flange lip) to where the tyre leaves the rim-flange lip round. At the TRA max-shoulder with the modelled section is Ø378.6, inside the TRA max Ø378.9.
 - Main hub-bearing (POH) bearings on the outer hubs, paired in the leg-door colour; no face stand 3 mm outward of the tyre's max width; retracted (wheel steel with the skin) & less 10.0 mm below the skin, inside the tyres 23.5 mm (POH-1, H, RL-check 5).
 - Main brake: six pistons, three retractors, steel friction surfaces, bolted to the axle (POH 7-4-10), 3 rotors / 2 stators inferred from the overload H1 (2 pairs, 40 wheels). Housing, stack and torque tube sized inside the inboard wheel half (E); one piston tube is shown in section.
 - No markings (owner decision): no sidewall lettering, brand names or placards on wheels or tyres. The tread grooves are geometry (the render-time groove shade is off, Scribe, valve and tie-bolt clacking is the drawn pose (the wheels turn); outboard units are mirror images. The main leg door (LD-1) is not shown.

PILATUS PC-12 PRO

WHEELS & TYRES - MAIN AND NOSE WHEEL ASSEMBLIES

DWG NO	REV	SCALE	SHEET	UNITS
PC12-L4W	A	1:3 / 1:2	A0 · 1 OF 1	mm
DRAWN				DATE
Claude - loftkit parametric model				202

RECONSTRUCTION FROM PUBLISHED DATA - NOT FOR CONSTRUCTION